

Project Initialization and Planning Phase

Date	4 July 2026
Team ID	SWTID-2026-7907
Project Title	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

EduGenie is an AI-powered learning assistant that uses Google Gemini to provide concept explanations, intelligent question answering, quiz generation, text summarization, and personalized learning support. The solution simplifies learning, saves students' time, and improves exam preparation through fast and accurate AI assistance.

Project Overview	
Objective	Develop an AI-powered learning assistant that helps students understand concepts, generate quizzes and summaries, answer academic questions, and provide personalized learning support using Google Gemini AI.
Scope	The system supports school, college, and competitive exam students by offering concept explanations, AI-powered question answering, quiz generation, text summarization, and personalized learning recommendations through a simple web interface.
Problem Statement	
Description	Students spend significant time searching across multiple learning resources, making it difficult to understand concepts quickly and prepare effectively for exams. Existing platforms often lack personalized AI assistance.
Impact	This results in reduced learning efficiency, increased confusion, inconsistent study materials, and lower academic performance.
Proposed Solution	
Approach	Integrate Google Gemini AI with a web-based learning platform to provide instant concept explanations, intelligent question answering, quiz generation, text summarization, and personalized learning recommendations.
Key Features	- AI-powered Question Answering - Concept Explanation - Quiz Generator

	- Text Summarization - Personalized Learning Path - Google Gemini Integration - Fast & Accurate Responses - User-Friendly Interface
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 / AMD Ryzen 5 Processor or Google Colab T4 GPU
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	FastAPI
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, size, format	Educational Notes (PDF), Student Questions, Quiz Dataset, Learning Content, User Learning History (PDF, TXT, CSV, JSON)